

Design and Implementation of Artificial Intelligence-Driven Network Intrusion Detection System

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Abstract—The current network intrusion detection systems (IDS) have poor adaptive learning capabilities under large-scale network traffic, resulting in low detection quality and efficiency. This article investigates the design and implementation of an artificial intelligence (AI)-driven network IDS to improve detection accuracy and real-time performance. Firstly, the one-hot encoding conversion method is used to denoise large-scale data. Then, a One-Class support vector machine (SVM) is used to optimize the adaptive learning and training process. Finally, real-time detection and updates are achieved by sliding the window to process data within a fixed time. The experimental results show that compared to the baseline systems of convolutional recurrent neural networks (RNN) and new deep neural networks (NDNN), the average accuracy of the system in this article is 4.7% and 6.5% higher, respectively, and the average response time decreases by 2.5 seconds and 4.1 seconds, respectively. The results indicate that AI-driven network IDS can provide more effective solutions for network security.

Keywords—network intrusion detection, artificial intelligence, one-class support vector machine, adaptive learning, system design and implementation.

I. INTRODUCTION

With the rapid development and popularization of Internet technology, the network has become an essential carrier of information transmission and exchange in modern society, and network security issues have become increasingly prominent. Network intrusion, as a malicious attack, threatens to personal privacy, trade secrets, and national security. It has become a significant challenge that urgently needs to be addressed [1]. With the increasing complexity and concealment of network attack methods, traditional intrusion detection systems (IDS) can no longer meet the protection needs of network security. Artificial intelligence (AI) has the characteristics of high flexibility and adaptive learning, which can automatically learn and extract complex feature representations, effectively responding to the diversity and variability of network intrusion. In the design of network IDS, the application of AI can improve the precision of network detection and enhance detection efficiency, which is of great value in meeting practical application needs.

This article studies the design and implementation of AI-driven network IDS and conducts experimental comparisons with baseline models to improve the efficiency and quality of network security detection. The results show that the average accuracy of the intrusion detection system in this article is 4.7% and 6.5% higher, respectively. From the real-time results, the AI-driven system in this article reduces the average response time to various attacks by 2.5 seconds and 4.1 seconds, respectively, compared to the baseline models. AI-driven network IDS can significantly improve detection

effectiveness and efficiency and help maintain network security.

This article describes the research background and significance of the IDS system in the first section. A review of current research is conducted in section 2. In section 3, research is conducted on designing and implementing network IDS driven by AI. The experimental results are analyzed and discussed in section 4. Section 5 summarizes the research findings, contributions, limitations, and prospects.

The research contribution of this article lies in the following:

- (1) Improved adaptive learning capability: this article proposes an AI-driven network IDS design and implementation that utilizes a combination of one-hot codes and One-Class support vector machines to enhance adaptability to unknown attack patterns so as to address the issue of traditional IDS being unable to handle massive data streams effectively.
- (2) Improved detection accuracy: the AI-based IDS in this paper has an average recognition rate of 89.2% for various network attacks, which is better than the existing convolutional RNN and NDNN advanced methods and can better identify and respond to network attacks.
- (3) Improved real-time performance: this article utilizes sliding window technology to process data streams in real time, improving the system's response speed. This improvement ensures that the system can respond to potential security incidents in the initial stage, thereby reducing losses.

II. RELATED WORK

With the increasing complexity of the network environment, personal information and data security issues caused by network intrusion are becoming more and more severe [2-3]. To protect computer networks and overcome network security issues, Sultana Nasrin researched on the application of Software Defined Networking (SDN) technology in network IDS, and the evaluation results showed that the technology could effectively detect and monitor network security issues caused by the emergence of programmable functions [4]. The extension of intelligent machine learning algorithms in network IDS through SDN has also attracted considerable attention. Alzahrani Abdulsalam O used machine learning algorithms for traffic monitoring to detect malicious behavior in the network as part of the network IDS in the SDN controller. Finally, its attack detection and classification accuracy reached 95.95% on the dataset [5]. Sakr Mahmoud M proposed an anomaly-based

network IDS that used SVM as a classifier for network connections and adjusted SVM control parameters using a standard particle swarm optimization algorithm. The acceptable evaluation results indicated that the characteristics of the proposed system could achieve high detection accuracy [6]. The current network IDS can resist attacks and risks to a certain extent, but most rely on pre-defined feature libraries to identify known malicious behavior, which has limitations in feature detection and classification of unknown threats.

With the maturity of AI technology, machine learning (ML) and deep learning (DL) algorithms provide more possibilities for adapting and identifying new threat features through automatic learning [7-8]. Gurung Sandeep used the DL method to design an IDS that detected known category attacks through pattern matching, applied sparse autoencoder for unsupervised feature learning, and used a logical classifier to classify the dataset. The results indicated the system had great potential for future applications [9]. Khan Muhammad Ashfaq created a hybrid IDS system using convolutional RNN, which captured local features through convolution and improved the performance and prediction ability of IDS by capturing temporal features through RNN. The simulation results demonstrated that the proposed method was significantly superior to current IDS, with a malicious attack detection accuracy of up to 97.75% [10]. Jia Yang designed an IDS based on an NDNN model, which modeled an NDNN with four hidden layers to capture and classify intrusion features. Experiments on datasets showed that the NDNN-based method improved the performance of IDS with an accuracy rate of up to 99.9% [11]. The current network IDS based on ML and DL algorithms provides adequate support for accurately identifying unknown threats, but its operational efficiency is low, especially when dealing with large-scale network traffic, and real-time detection is challenging to meet application requirements.

III. METHODS

A Demand Analysis

IDS is a device or software used in modern organizations to resist attacks, sabotage, and other functions. It is a security system that can monitor object files in real time. Once any abnormalities are detected, it can promptly sound an alarm. As a forward-looking security protection technology, IDS can effectively identify attacks. When an intrusion occurs, it actively responds, intercepts it, and collects evidence of the intrusion.

In abnormal intrusion detection, abnormal behavior includes attack behavior, and theoretically, abnormal behavior is called an attack. However, abnormal behavior does not necessarily equate to invasion. It mainly consists of four situations: (1) no abnormal intrusion behavior; (2) non-invasive abnormal behavior; (3) neither invasive nor abnormal behavior; (4) simultaneously exhibiting invasive and abnormal behavior. Abnormal intrusion detection refers to the analysis and modeling of constructed abnormal behaviors. Abnormal intrusion detection is the use of known data biases to predict user behavior and make judgments about it.

Based on its practical application characteristics, this article analyzes it from four dimensions: data collection, intelligent detection and analysis, real-time monitoring and alarm, and user interface and reporting:

1) *Data collection*: The system requires strong data acquisition capabilities and the ability to collect real-time data from various sources, such as network traffic, system logs, application logs, etc. The data collection module should have high availability and scalability to support multiple data formats and protocols to ensure data integrity and real-time performance.

2) *Intelligent detection and analysis*: The system should support intelligent detection and analysis to select the most reflective features of intrusion behavior from massive data. At the same time, it should have the function of automatically screening and optimizing models to choose the optimal model based on the characteristics and actual needs of the data.

3) *Real-time monitoring and alarm*: The system should have a real-time monitoring function, which can process and analyze network traffic and logs in real-time and detect possible attacks promptly. The design of the alarm mechanism should ensure both the reliability of the system and the quick and efficient response of users to attacks without causing any impact on the normal service of users. The security measures for system automation must include blocking malicious data streams, isolating infected hosts, triggering alarms, etc.

4) *User interface and reporting*: The system should also have a good human-computer interaction interface to facilitate the operation and monitoring of the system by security analysts and management personnel. The user interface should include a data visualization module that can present network traffic, system logs, and test results through graphics and dashboards. The report generation module needs to provide various forms of output such as PDF, Excel, HTML, etc., for users to easily store and share.

B System Architecture

Figure 1 shows the overall architecture of AI-driven network IDS:

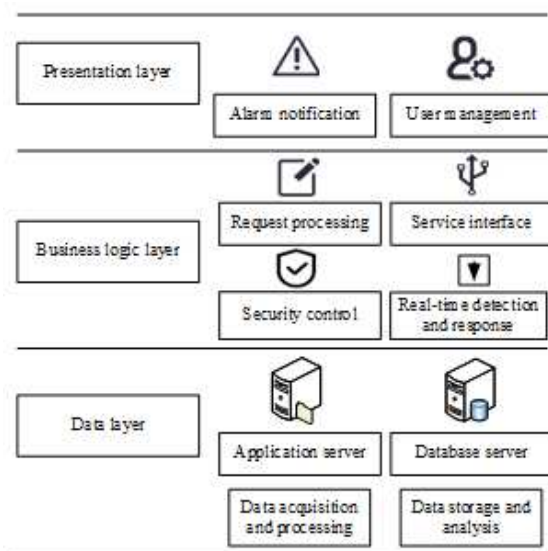


Fig 1. Overall system architecture

Figure 1 shows that the overall architecture of AI-driven network IDS consists of a data layer, a business logic layer, and a presentation layer.

The presentation layer is the interface for user-system interaction and is mainly responsible for displaying and receiving user instructions. At the presentation level, users can see real-time network conditions through alarm notifications and user management. In the presentation layer, front-end technology is applied and combined with the React framework to enable dynamic data interaction. WebSocket technology is utilized to achieve effective communication between the presentation layer and the business logic layer, ensuring that data in the interface can be updated promptly, thereby improving the system's response speed and user experience.

The functions of the business logic layer include request processing, service interface, security control, and real-time detection and response. AI technology models network traffic data and discovers possible attack methods. This layer obtains historical data and real-time traffic information through interaction with the data layer, and responds quickly to various new attack patterns.

The data layer consists of application servers and database servers, responsible for collecting, storing, and maintaining network business and logs. Through collection devices, network traffic is collected in real time and stored in a database. By adopting data warehouse technology, massive amounts of data are integrated and analyzed.

C Hardware Design

1) *Network traffic collection device*: The network traffic collection device consists of network interfaces, deep packet inspection (DPI), and processing units. Collection devices are configured on edge routing network nodes to monitor all incoming and outgoing packets in real time. Business features are extracted, including source address, destination address, and protocol type. The collected data is transmitted to the business logic layer through a secure channel to achieve rapid identification and response to network threats.

2) *Detection servers*: The detection server adopts multi-core processors and large-capacity memory. In terms of storage, a combination of solid-state storage and hard disk drives is adopted. Based on pre-trained data-driven models, real-time analysis of potential intrusion behaviors is achieved through data cleaning and feature extraction. When abnormal behavior is detected, the system generates an alarm message and transmits it to the business logic layer.

3) *Controller*: The controller consists of a task scheduling component and a data coordination component. Firstly, requests are received from the front end, and the parameters of the requests are parsed. Then, the back-end services are scheduled according to the type of request. The configuration of each task is optimized through task scheduling components, thereby achieving the allocation of computing resources for each task. The data coordination component coordinates the data flow between different layers. Once abnormal data is received, the alarm mechanism may be quickly activated, and messages may be pushed to the user interface.

D Implementation of AI-driven Intrusion Detection

After obtaining the user's network traffic data, in order not to affect the accuracy of intrusion detection, the system uses a one-hot code transformation method to extract digital features instead of character features in the original network data and

uses a fixed-size state register to denoise massive data. A new feature set is obtained by expanding the original data's features and removing unnecessary features.

Using the Min-Max normalization method, the encoded feature values are further processed to ensure that all features are within the range of [0,1]:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (1)$$

x is the original feature, and x' is the normalized feature value.

After normalization, the preprocessed user network traffic data is used as the input vector, and One-Class SVM is used as the basic learning mode. It is assumed that $x_1, x_2, \dots, x_n$ are the training samples, representing standard class data points. The One-Class SVM optimization problem can be expressed as [12-13]:

$$\begin{aligned} \min_{\omega, b, \xi} & \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} & \omega^T \phi(x_i) + b \geq 1 - \xi_i, \xi_i \geq 0, i=1, 2, \dots, n \end{aligned} \quad (2)$$

Among them, Table 1 shows the definitions of variables:

TABLE I. DEFINITION OF FORMULA VARIABLES

Sequence	Variables	Meaning
1	ω	Weight vector
2	b	Bias direction
3	$\phi(x_i)$	Mapping function
4	ξ_i	Slack variable
5	C	Punishment coefficient

In large-scale data, samples with multiple information features are selected from the data source and are annotated to avoid extracting individual samples one by one:

$$M = \begin{cases} \text{Labeled} = \langle p_1 | m \rangle - \langle p_2 | m \rangle \leq \theta \\ \text{otherwise, unlabeled} \end{cases} \quad (3)$$

Among them, M represents whether the sample needs to be labeled, and p_1, p_2 represent attribute values.

It is necessary to be able to process data quickly to meet the requirements of real-time intrusion detection. This article uses a sliding window method to process and update the collected data regularly. Assuming the size of the sliding window is T , at each time t , the parameter θ of the model is updated:

$$\theta_{t+1} = \theta_t + \eta (\nabla L(\theta_t; x_{t-T:t}) - \nabla R(\theta_t)) \quad (4)$$

Among them, η is the learning rate; L is the damage function; R is the regularization term.

On this basis, utilizing annotations in the data source, rapid and accurate identification of labeled and unlabeled abnormal data is achieved to achieve real-time detection and processing of intrusion threats.

IV. RESULTS AND DISCUSSION

A. Experimental Data

This article evaluates the detection accuracy and real-time performance separately to verify the effectiveness of AI-driven network IDS. It is compared with the convolutional

RNN shown in reference 10 and the NDNN baseline system shown in reference 11 to test the system's performance in this article objectively. For the convenience of system testing, this article uses the CIC (Canadian Institute for Cybersecurity) FlowMeter network security simulation tool to generate simulated attack traffic. Table 2 shows the types and quantities of attacks.

TABLE II. ATTACK TYPES AND QUANTITIES

Sequence	Amount	Specific type
1	108	Ping of death
2	89	Phishing
3	135	Zero-day
4	194	Advanced persistent threat
5	106	Cross-site scripting
6	121	Password attack
7	233	Cryptojacking

B. Detection Accuracy

Regarding detection accuracy, 10 repeated experiments are conducted for each type of attack to compare and analyze the average accuracy of various attack events in different systems. Figure 2 shows the results:

The specific comparison results in Figure 2 show that the system in this article has an average accuracy of 89.2% for various types of attacks. The system based on convolutional RNN achieves an average accuracy of 84.5%. The system based on NDNN achieves an average accuracy of 82.7%. Compared to the baseline systems, the average accuracy of intrusion detection in this article's system is 4.7% and 6.5% higher, respectively. Through adaptive learning, this article's system is based on One-Class SVM and adopts unsupervised adaptive learning methods to mine more features from large-scale traffic data, characterize more dimensional information and improve detection accuracy.

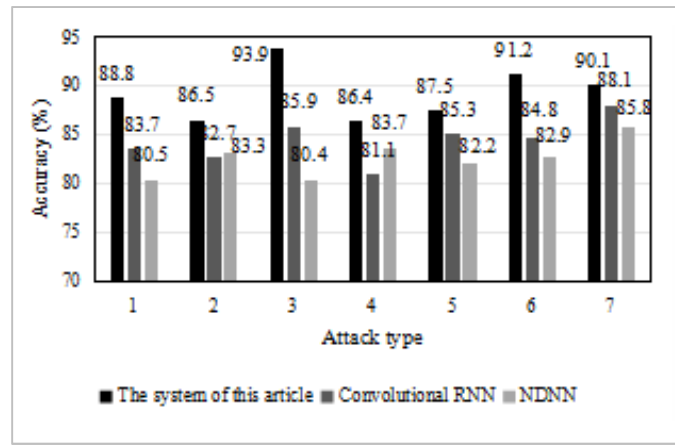


Fig 2. Comparison of detection accuracy

C. Real-time Performance

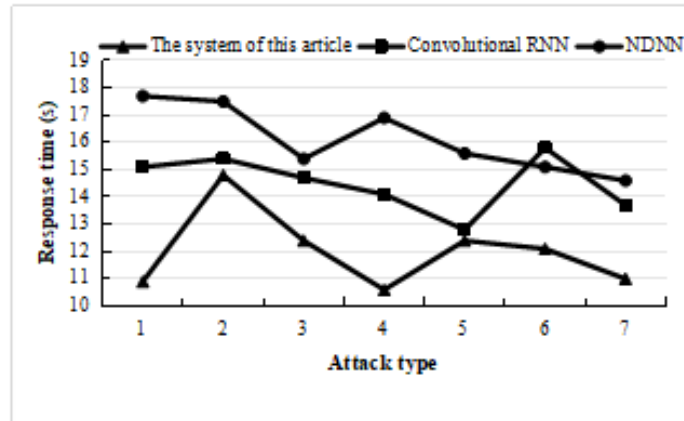


Fig 3. Real-time performance comparison

In real-time comparison, 10 repeated tests are conducted on network attack behavior, and then the average response time of the three types of systems is compared, as shown in Figure 3.

The real-time comparison of Figure 3 shows a significant difference in response time among the three types of systems. From the specific comparison results, in repeated testing, the average response time of the system in this article to various attacks is about 12.0 seconds. The average response time of

baseline systems based on convolutional RNN and NDNN is 14.5 seconds and 16.1 seconds, respectively. Compared to these two types of systems, the AI-driven system in this article reduces the average response time to various attacks by 2.5 seconds and 4.1 seconds, respectively. Through effective data preprocessing, this system effectively avoids the impact of noise on real-time detection. It updates the training process in real-time through sliding windows, effectively improving data processing speed and real-time performance of detection.

D. Discussion

1) *Result impact*: This article presents the design and implementation of AI-driven IDS. By using techniques such as one-hot encoding, One-Class support vector machine, and sliding window, the detection accuracy is improved to 89.2%. At the same time, the average response time is shortened to 12 seconds. These results indicate that AI-driven IDS can effectively extract features from massive data and adapt to new attack patterns in real time, thereby enhancing network security.

2) *Comparative research analysis*: The experimental results show that this IDS has high recognition accuracy and fast response capability. Compared to systems based on convolutional RNN (84.5%) and NDNN (82.7%), the average accuracy of the system in this article is 4.7% and 6.5% higher, respectively. Regarding real-time performance, the average response time when facing attacks is also reduced by 2.5 seconds and 4.1 seconds, respectively. This article combines the unsupervised learning of SVM and the fast updating of sliding window technology to better adapt to unknown threats.

3) *Overall analysis of results*: Although this article provides some guidance for improving network security, it also has limitations. Because the experimental environment is based on simulated flow and lacks comprehensive evaluation of actual data, further validation is needed for its application in a wide range of practical scenarios. In addition, although CIC FlowMeter is used to generate attack flows, in real network environments, attack behaviors are more complex and varied, requiring higher performance metrics. In future research, it can be considered to optimize the algorithm in complex and dynamic network environments and, on this basis, improve the generalization ability and robustness of the model by applying high-dimensional data to achieve higher intrusion detection performance.

V. CONCLUSIONS

In the context of the complexity of the network environment, IDS has become an increasingly important means of maintaining information and asset security. This article studies the design and implementation of AI-driven IDS to improve detection efficiency and quality and enhance personal data security. Through unsupervised adaptive learning and window updates, it effectively improves the detection accuracy of network attacks and enhances data processing speed and real-time detection. Although this study can improve the system detection performance to a certain extent and provide guidance for data information security, there are still limitations. This article does not consider the study of system robustness in dynamically changing network environments. In future development, it is necessary to consider the impact of network traffic dynamics on system detection performance to continuously improve system

performance and promote high-quality development of network security.

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